# Advocate.NOW

# ABSTRACT

The project titled "AI Lawyer - Legal Advice Chatbot using NLP" aims to develop an intelligent legal assistance system that uses Artificial Intelligence (AI) and Natural Language Processing (NLP) to provide quick, reliable, and accessible legal information to users.

This system acts as a virtual legal advisor where users can ask questions in plain English and receive accurate, context-aware answers. It eliminates the need for manual consultation for basic legal queries and helps bridge the gap between people and legal professionals.

The project operates through three main modules - Admin, Lawyer, and User.

-The Admin module manages the overall system, verifies lawyer details, handles complaints, updates old case records, and ensures smooth functioning of the platform.

-The Lawyer module allows verified advocates to register, manage appointments, update case and hearing details, and interact with users.

-The User module provides an interactive chatbot interface for general legal queries, booking lawyer appointments, submitting complaints, and viewing case information.

By integrating NLP, the chatbot understands and responds intelligently to human queries. Additionally, the system includes features like old case reference access, lawyer verification, and police station directory, enhancing both trust and transparency.

This AI-based platform significantly reduces time, effort, and cost in obtaining legal help while promoting awareness and accessibility of legal resources. It demonstrates how technology can be effectively utilized to make legal assistance more efficient, transparent, and user-friendly.

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# 1. INTRODUCTION

*Advocate.NOW* is an AI-powered Lawyer Chatbot and Appointment Booking System designed to simplify access to legal assistance through intelligent automation and digital scheduling. The platform aims to bridge the gap between clients and legal professionals by providing instant legal guidance, efficient lawyer discovery, and seamless appointment booking, ensuring a smooth and user-friendly experience for all stakeholders.

The system integrates web-based modules for Admin, Lawyers, and Clients, along with an advanced AI chatbot that assists users by answering common legal questions, offering preliminary legal insights, and guiding users toward appropriate legal services. By leveraging artificial intelligence, *Advocate.NOW* minimizes the need for manual consultations for basic legal queries while improving response time and accessibility.

The Lawyer Module provides legal professionals with tools to manage their profiles, specify areas of expertise, set availability, accept or decline appointments, and communicate securely with clients.

The Client Module ensures a convenient and intuitive experience by offering features such as easy registration, interaction with the AI lawyer chatbot, browsing verified lawyers, booking appointments, secure payments, tracking appointment status, and submitting reviews or complaints. The AI chatbot acts as the first point of contact, helping users understand their legal concerns before connecting them with a suitable advocate.

By automating legal assistance and appointment scheduling, Advocate.NOW enhances operational efficiency, improves communication between clients and lawyers, and increases accessibility to legal services. The system contributes to a well-structured, transparent, and efficient legal service ecosystem, ultimately improving user satisfaction and productivity.

# 2. SYSTEM ANALYSIS

**2.1 IDENTIFICATION OF NEED**

*Advocate.NOW* fulfills this need by integrating an AI lawyer chatbot with an online appointment booking system. The platform enhances accessibility, improves efficiency, and ensures transparency in legal services. By digitizing legal assistance and automating routine processes, *Advocate.NOW* addresses the limitations of traditional legal consultation systems and meets the increasing demand for smart, user-friendly legal solutions.

Traditional legal consultation methods rely heavily on physical visits, manual appointment scheduling, and direct communication with lawyers, which can be inefficient and inconvenient. Clients often struggle to find suitable advocates based on their legal needs, availability, and expertise. Additionally, lawyers face difficulties in managing appointments, handling repetitive client queries, and maintaining organized communication, which impacts productivity and service quality.

## 2.2 PRELIMINARY INVESTIGATION

The preliminary investigation for the *Advocate.NOW* project was conducted to understand the existing legal consultation systems, identify their limitations, and assess the feasibility of integrating artificial intelligence into legal service delivery. This investigation involved studying traditional legal consultation methods, online legal service platforms, and emerging AI-based legal assistance tools.

The study revealed that most conventional legal service systems rely on manual processes such as physical visits, phone calls, and unstructured appointment scheduling. These methods often result in long waiting times, lack of availability information, and inefficient communication between clients and lawyers. Additionally, many existing online platforms primarily focus on listing lawyers and booking appointments, offering little to no instant legal guidance for users.

During the investigation, it was observed that users frequently seek basic legal information before deciding to consult a lawyer. However, the absence of instant guidance forces users to either rely on unreliable online sources or schedule unnecessary consultations, increasing cost and time.

### 2.2.1 FACT FINDING TECHNIQUES

Fact finding is a crucial step in system development, as it helps in understanding user requirements, existing system limitations, and functional expectations. For the *Advocate.NOW* project, multiple fact finding techniques were employed to collect accurate and relevant information from various stakeholders involved in the legal service ecosystem.

1. **Interviews**

Structured and semi-structured interviews were conducted with legal professionals and potential users to understand their challenges, expectations, and workflow. Lawyers provided insights into appointment management issues, repetitive client queries, and communication gaps, while users shared their difficulties in accessing timely legal advice and scheduling consultations.

2. **Questionnaires**

Online questionnaires were distributed to a broader audience, including students, working professionals, and small business owners. The questionnaires helped gather quantitative data regarding user awareness of legal processes, interest in AI-based legal assistance, preferred appointment modes, and concerns related to privacy and reliability.

3. **Observation**

The existing legal consultation process was observed to understand how appointments are booked, how consultations are conducted, and how records are maintained. This technique helped identify inefficiencies such as manual record-keeping, lack of scheduling transparency, and dependency on physical presence.

**4. Document Analysis**

Relevant documents such as legal service workflows, existing online legal platforms, research papers, and case studies on AI chatbots were analysed. This helped in understanding industry standards, legal service requirements, and the feasibility of implementing AI-based legal assistance.

**5. Internet Research**

Extensive internet research was carried out to study existing legal tech solutions, AI chatbot frameworks, security practices, and user experience standards. This provided insights into current trends, best practices, and technological possibilities that could be incorporated into *Advocate.NOW*.

## 2.2.2 Existing System

The existing legal consultation system primarily operates through traditional, manual, and semi-digital methods. Clients seeking legal assistance usually rely on physical visits to law offices, phone calls, or personal referrals to contact advocates. Appointment scheduling is often handled manually, leading to inefficiencies such as long waiting times, scheduling conflicts, and lack of real-time availability information.

In the current system, clients must directly consult a lawyer even for basic legal queries, which can be time-consuming and costly. There is no provision for instant preliminary legal guidance, forcing users to depend on unreliable online sources or informal advice. This lack of structured initial support creates confusion and delays in addressing legal issues.

Existing online legal platforms mainly focus on listing lawyers and providing appointment booking features. However, most of these platforms do not offer intelligent legal assistance or personalized guidance based on user queries. Communication between lawyers and clients is often unstructured, and follow-ups are handled through external messaging or email systems, resulting in poor record management.

## 2.2.3 Proposed System

*Advocate.NOW* is an AI-powered Lawyer Chatbot and Appointment Booking platform designed to overcome the limitations of the existing legal consultation system. The system provides a unified digital solution that offers instant legal assistance, efficient appointment management, and structured communication between clients and advocates.

*Advocate.NOW* introduces an intelligent AI lawyer chatbot that acts as the first point of interaction for users. The chatbot provides preliminary legal guidance, answers frequently asked legal questions, and helps users understand their legal concerns before consulting a lawyer. This reduces unnecessary consultations and saves time for both clients and legal professionals.

• **Admin Module:** Manages advocate registration and verification, monitors system activities, oversees appointments, handles user reviews and complaints, and ensures smooth and secure functioning of the *Advocate.NOW* platform.

• **Lawyer Module:** Allows lawyers to manage their profiles, specify areas of legal expertise, set availability, view and manage appointment requests, communicate with clients, and track consultation history through an organized digital interface.

• **User Module:** Enables users to register and log in, interact with the AI lawyer chatbot, browse verified lawyers, book appointments, make secure payments, track appointment status, and submit feedback or complaints.

## 2.3. FEASIBILITY STUDY

A feasibility study was conducted to evaluate the practicality and viability of developing and implementing the *Advocate.NOW* system. The study analyses the system from technical, economic, operational, and legal perspectives to ensure that the proposed solution is realistic and sustainable.

### 2.3.1. Technical Feasibility

The *Advocate.NOW* system is technically feasible due to the availability of modern web and mobile technologies, artificial intelligence frameworks, and secure cloud infrastructure. The AI chatbot can be developed using natural language processing techniques, while appointment booking, user authentication, and payment processing can be implemented using existing frameworks and APIs. The required hardware and

software resources are readily available, and the development team possesses the necessary technical skills to implement the system efficiently.

**Our system's technical requirements include:**

• **Frontend Technologies:** HTML, CSS, JavaScript for creating responsive and user-friendly web interfaces.

• **Backend Development:** Python (Django) in PyCharm for robust server-side logic, API development, and database management.

• **User Application:** Developed using Android Studio with Flutter, providing a smooth and native experience for mobile users.

• **Database Management:** SQL Yog with MySQL for efficient data storage, retrieval, and management.

• **Operating System:** macOS for development and deployment environments.

### 2.3.2. Economic Feasibility

The development cost of *Advocate.NOW* is relatively low as it primarily relies on open-source technologies and cloud-based services. Automation of legal assistance and appointment scheduling reduces operational costs for legal professionals and minimizes administrative overhead. The system can be scaled based on demand, making it economically viable for long-term use. Overall, the benefits of improved efficiency and reduced manual effort outweigh the development and maintenance costs.

# 3. SOFTWARE REQUIREMENT SPECIFICATION

## 3.1. FUNCTIONAL REQUIREMENTS

A functional requirement describes what a software system should do.

Functional requirements define the system's functionalities, behaviour, and objectives.

**The *Advocate.NOW* system has the following functional requirements:**

| Functional Requirements | Importance | Description |
|---|---|---|
| Security | Essential | User login is required |
| Web interface | Essential | Provides an intuitive and interactive web interface for Admin and Lawyer modules. |
| Database | Essential | Stores user details, lawyer profiles, appointment information, chatbot interactions |
| AI Lawyer Chatbot | Essential | Provides instant preliminary legal guidance by answering user queries using artificial intelligence. |
| Appointment Booking System | Essential | Allows users to book legal consultations and view lawyer availability in real time. |
| Lawyer Management | Important | Enables lawyers to manage profiles, expertise areas, availability, and appointment requests. |
| User Management | Important | Allows users to register, log in, manage profiles, and track appointment history. |

## 3.2. NON FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes of a system rather than specific functionalities. They describe how the system performs its operations and help achieve performance, reliability, and usability goals.

**The non-functional requirements of *Advocate.NOW* are listed below:**

| Non-Functional Requirement | Importance | Description |
|---|---|---|
| Performance | Performance | The system should provide fast response times for chatbot interactions, appointment booking. |
| User friendly environment | Essential | The system interface should be simple, intuitive, and easy to use for Admins, Lawyers, and Users. |
| Reliability | Essential | The system should operate consistently without failures and ensure accurate data processing. |
| Application Maintenance | Essential | System should be flexible for future use |
| Platform independence | Essential | System should be capable to work in any environment |

## 3.3. SYSTEM CONFIGURATION

The hardware and software configurations required for the installation and smooth functioning of *Advocate.NOW* are based on the operational needs of the system. Proper configuration ensures reliable performance, efficient processing, and compatibility across all modules, including Admin, Lawyer, and User applications.

**It includes two phases:**

### 3.3.1. HARDWARE CONFIGURATION

The selection of appropriate hardware is essential for the effective functioning of the *Advocate.NOW* system. **The minimum hardware requirements are as follows:**

**Processor** : Intel Core i3 / Apple silicon

**Primary Memory** : 8GB RAM and above

**Storage** : 256 GB SSD and above

**Display** : VGA Color Monitor

**Key Board** : compatible

**Mouse** : compatible

### 3.3.2. SOFTWARE CONFIGURATION

One of the most difficult task is selecting software for the system, once the system requirements is found out then we have to determine whether a particular software package fits for those system requirements. **The application requirement**:

**Operating System** : Windows / macOS

**Front End** : HTML, CSS, JavaScript

**Back End** : MySQL (Database) using SQL Yog

**Framework** : Django

**Script** : Python 3.6 or above

U**ser Application** : Dart (Flutter) in Android Studio

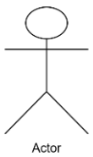
## 3.4. CONCEPTUAL MODELS

### 3.4.1. USE CASE DIAGRAM

A use case diagram is a graphic depiction of the interactions among the elements of a system. A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. In this context, the term "system" refers to something being developed or operated. Such as a mail-order product sale and service web site. Use case diagrams are employed UML (Unified Modeling Language), standard notation for the modeling of real world objects and system.

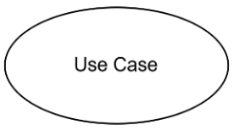
#### Actor

An actor represents a role that an outsider likes on when interactive with business system. For instance, an actor can be a customer, a business partner or another business system. Every actor has a name:




#### Use case

A use case represents a user goal that can be achieved by accessing the software application. In visual paradigm, you can make use of the sub-diagram feature to describe the interaction between user and system within a use case by creating a sub-sequence diagram under a use case. You can also describe the use case scenario using the flow of Events editor.



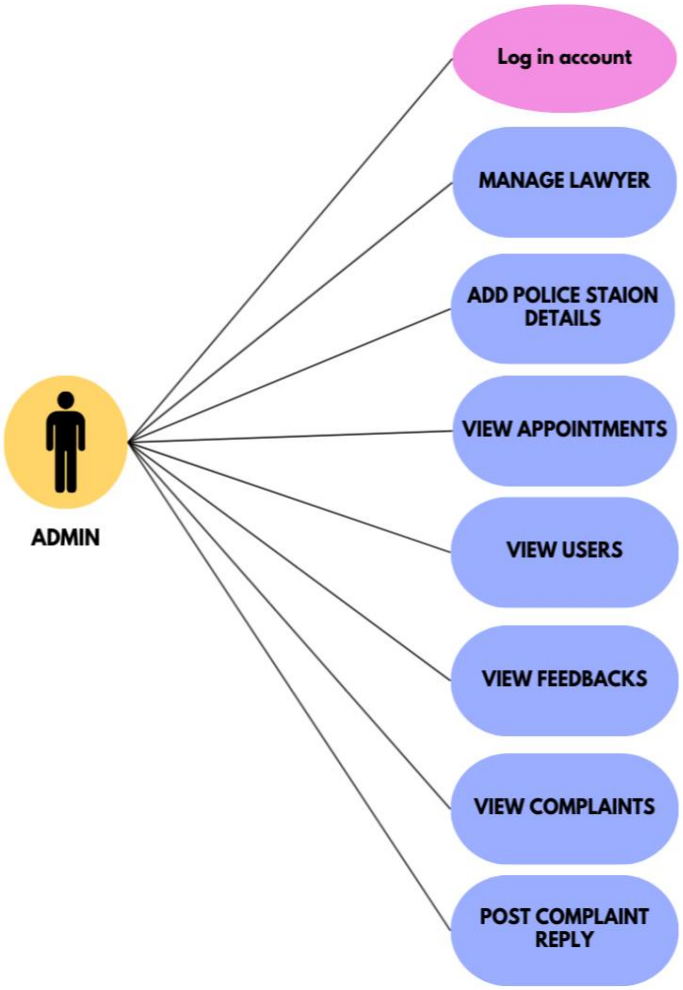

#### Relationships

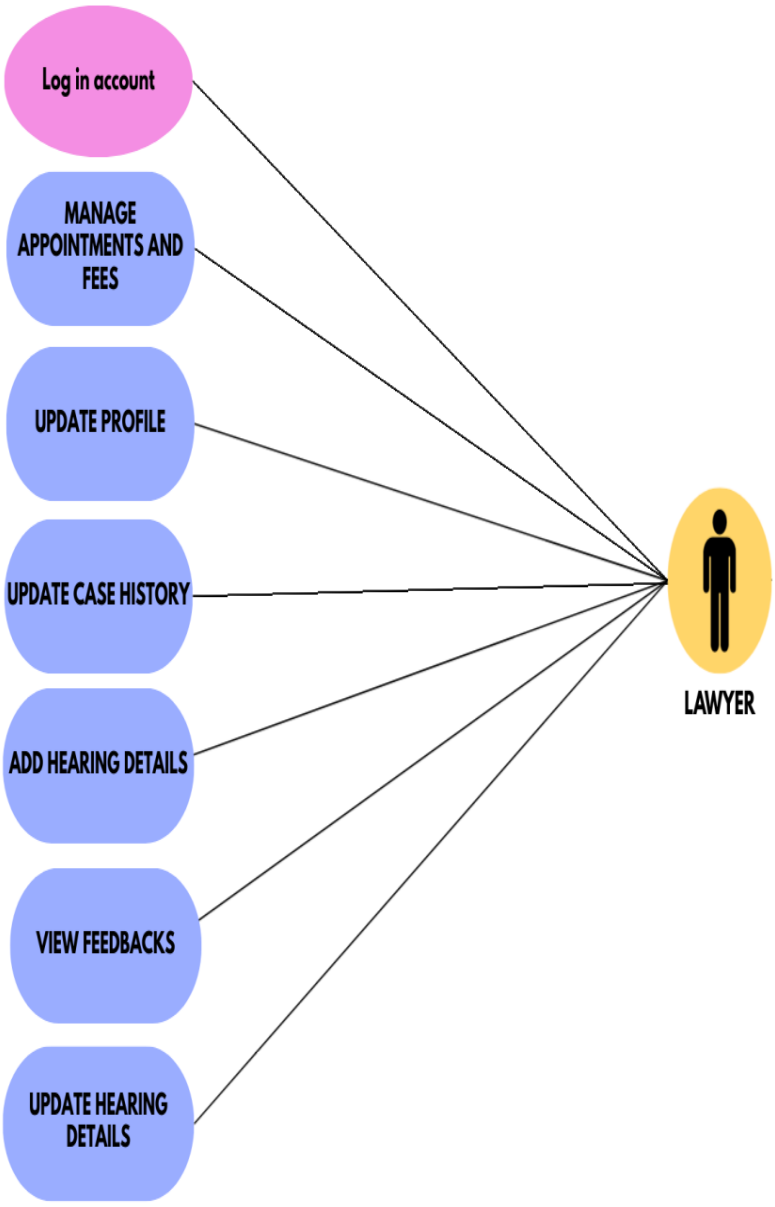
A line between actor and use case represents communication Typically, bidirectional (e.g. actor initiate the use case, a use case sends information to an actor)

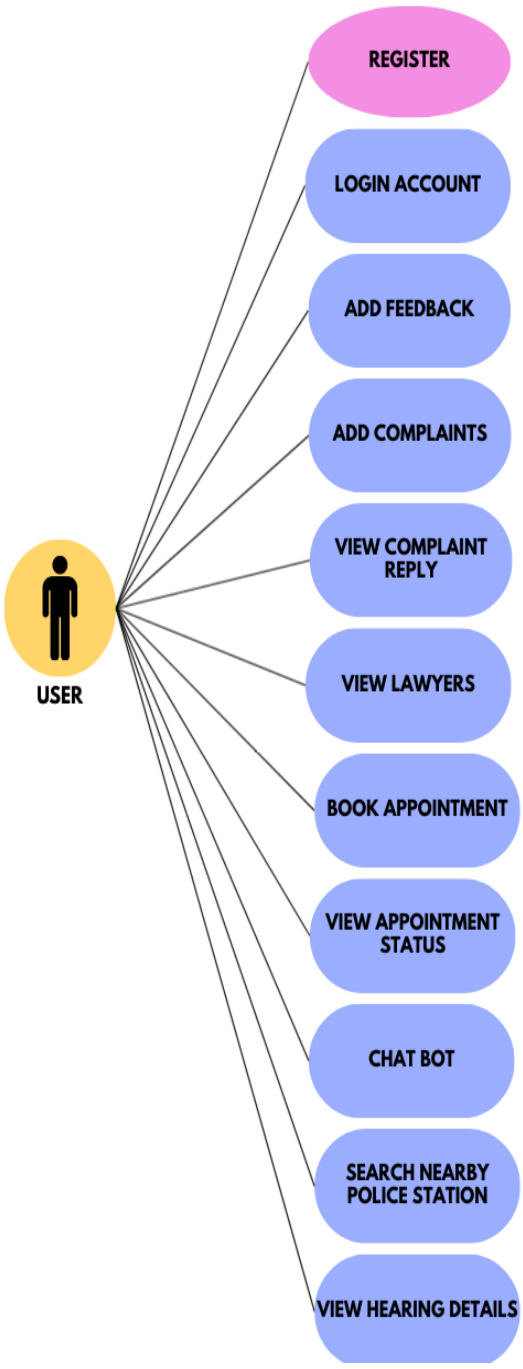
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## Use case Diagram :




Log in account
MANAGE APPOINTMENTS AND FEES
UPDATE PROFILE
UPDATE CASE HISTORY
ADD HEARING DETAILS
VIEW FEEDBACKS
UPDATE HEARING DETAILS
LAWYER


REGISTER
LOGIN ACCOUNT
ADD FEEDBACK
ADD COMPLAINTS
VIEW COMPLAINT REPLY
VIEW LAWYERS
BOOK APPOINTMENT
VIEW APPOINTMENT STATUS
CHAT BOT
SEARCH NEARBY POLICE STATION
VIEW HEARING DETAILS
USER

## 3.4.2. ACTIVITY DIAGRAM

An activity diagram is a graphical representation of the workflow of activities within a system. It is used to model the dynamic behavior of the system by showing the sequence of actions, control flow, and decision-making steps involved in a process. Activity diagrams are especially useful for describing business processes, use case workflows, and functional logic of the system.

In this context, an activity diagram illustrates how different activities are performed in a specific order from the start of a process to its end. It also supports modeling of parallel activities and conditional flows. Activity diagrams are part of UML (Unified Modeling Language), which is a standard notation used for modeling real-world systems and software behavior.

### Initial Node

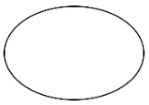
The initial node represents the starting point of an activity diagram. It indicates where the workflow begins and is represented by a filled black circle.



**Initial Node**

### Activity / Action

An activity represents a specific task or operation performed in the system. It is shown as a rounded rectangle and describes a step in the workflow.



**Activity**

### Control Flow

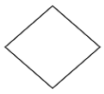
Control flow shows the order in which activities are executed. It is represented by arrows that connect activities and indicate the direction of flow from one activity to another.



**Control Flow**

## Decision Node

A decision node is used to represent a conditional branch in the workflow. It is shown as a diamond shape, where one incoming flow splits into multiple outgoing flows based on conditions.



**Decision Node**

## Fork and Join

A fork node is used to divide one flow into multiple parallel flows, allowing activities to occur simultaneously.

A join node synchronizes these parallel flows back into a single flow. Both are represented by thick horizontal or vertical bars.



**Fork and Join**

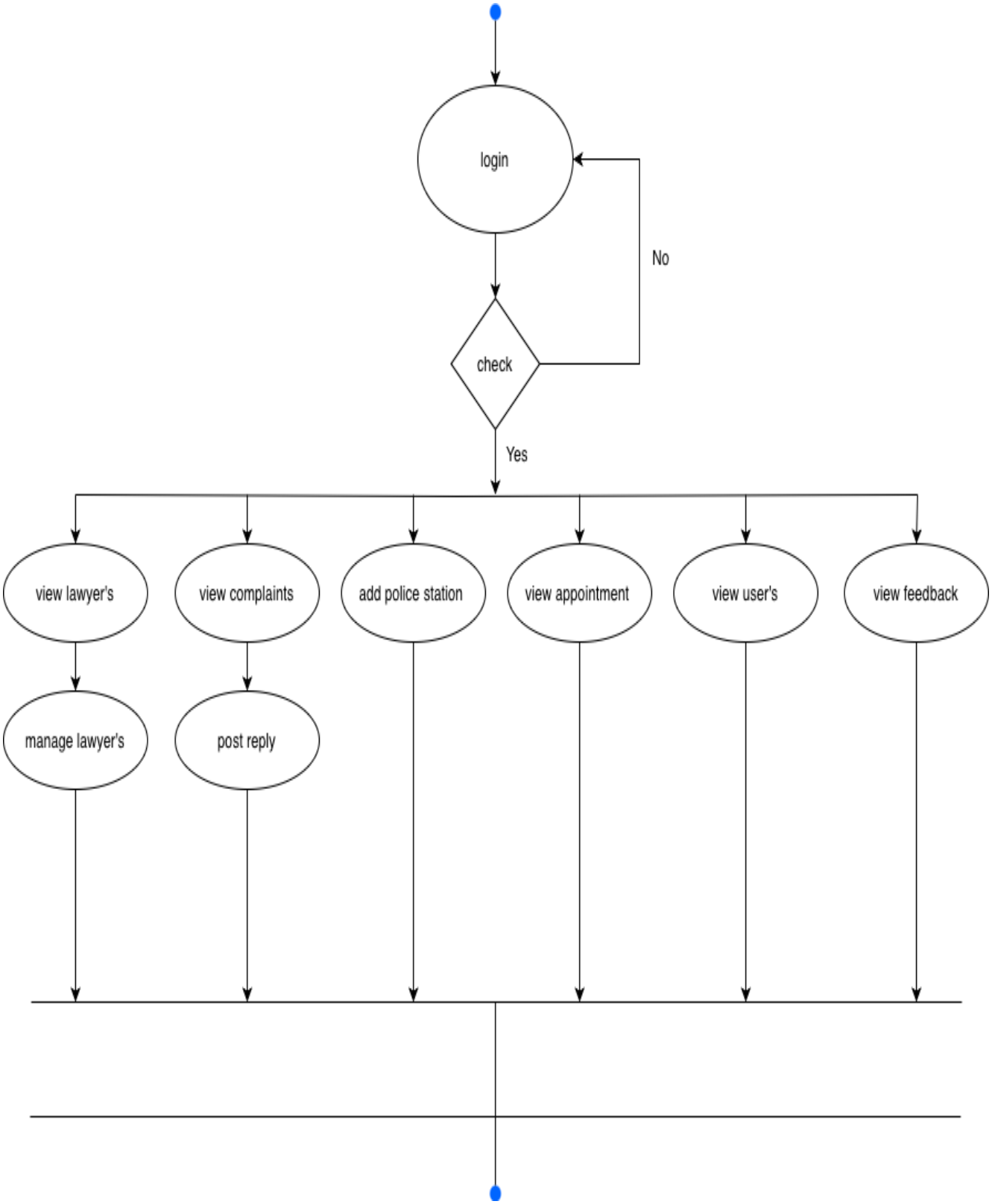
## Final Node

The final node represents the end of the activity flow. It indicates that the process has been completed and is shown as a black circle surrounded by an outer ring.

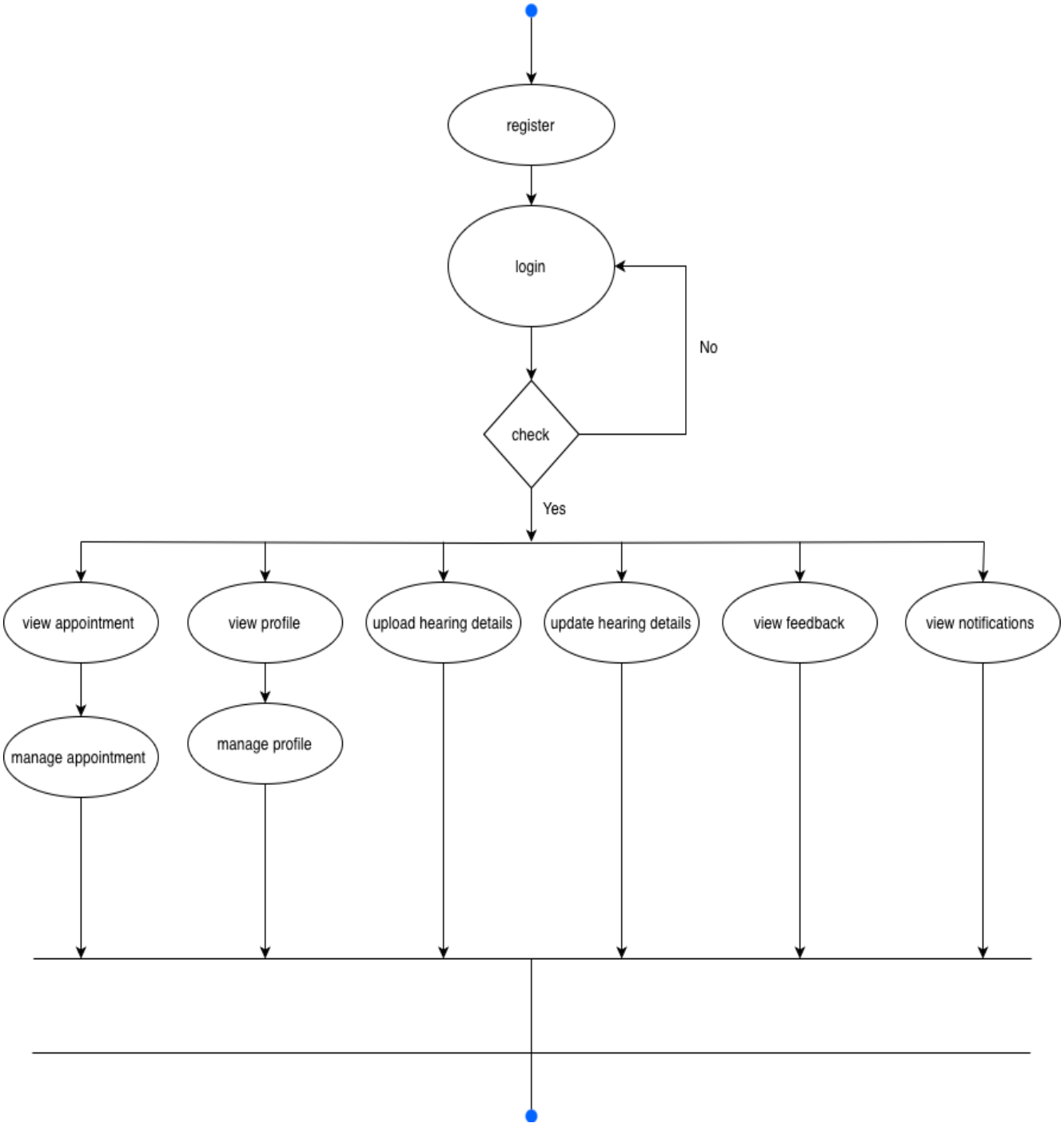


**Final node**

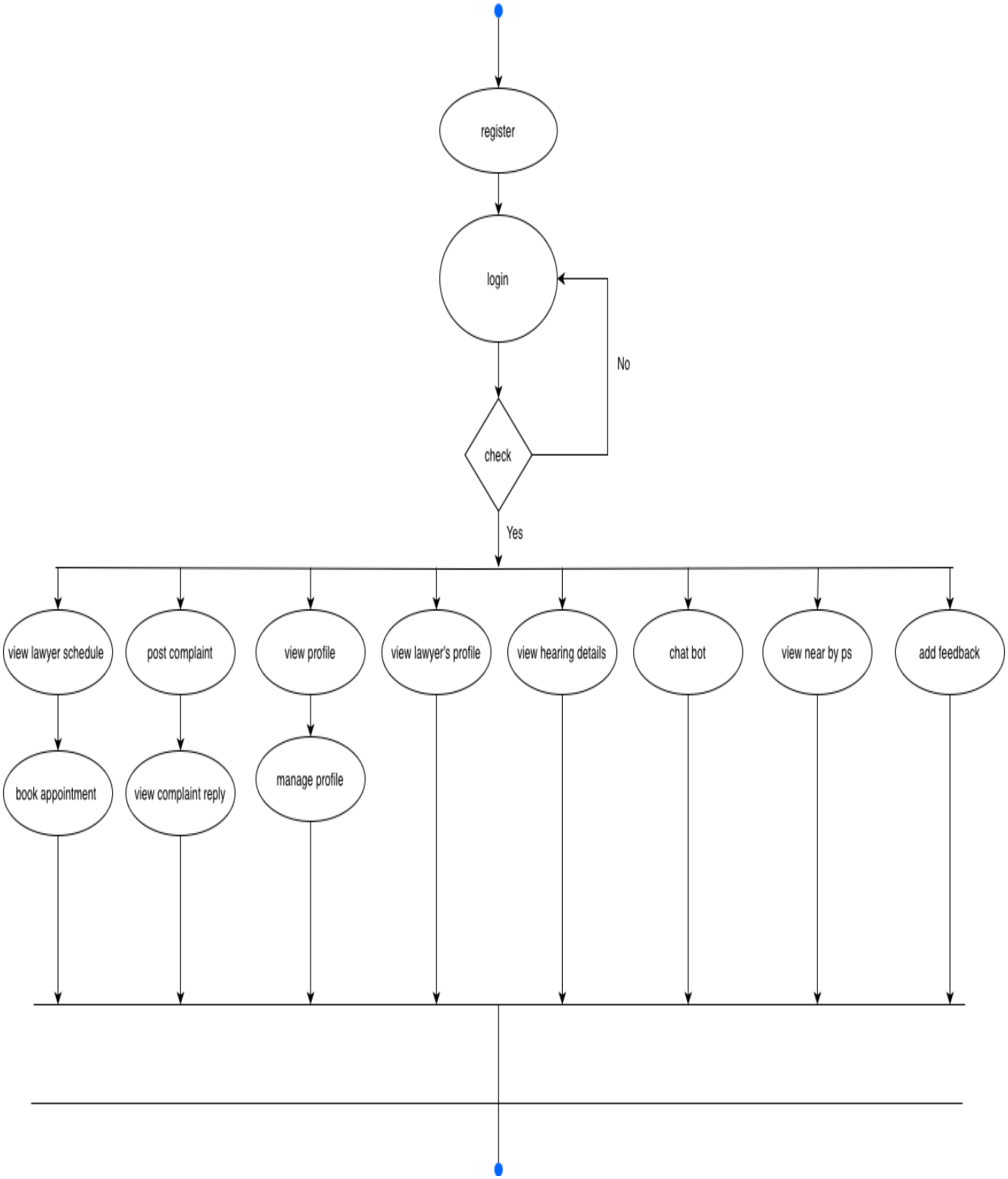
## ADMIN'S ACTIVITY DIAGRAM



## LAWYER'S ACTIVITY DIAGRAM



## USER'S ACTIVITY DIAGRAM



### 3.4.3. DATA FLOW DIAGRAM